



Faculty of computer science and information technology

Year 2018/2019

Level: 1

Course Name: Calculus 1

Course Code: MAT102

Sheet No.: 5

Semester: Fall 2018

Instructor: Dr. Rawhy Ismail

Evaluate the following limits:

a) $\lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{x + x^2} \right)$

b) $\lim_{x \rightarrow \infty} \left(\frac{5x^2 + 3x - 7}{2x} \right)$

c) $\lim_{x \rightarrow \infty} \sqrt{\frac{8x^2 + 7}{2x^2 + 5x - 1}}$

d) $\lim_{x \rightarrow \pi/2} \left(\frac{\sec x}{1 + \tan x} \right)$

e) $\lim_{x \rightarrow \infty} \left(\frac{4^x + 1}{3^x - 1} \right)$

f) $\lim_{x \rightarrow \infty} \left(\frac{\ln x}{2\sqrt{x}} \right)$

g) $\lim_{x \rightarrow 0^+} \left(\frac{\ln(x^2 + 2x)}{\ln x} \right)$

h) $\lim_{x \rightarrow \infty} \left(\frac{x - \sin x}{x \tan x} \right)$

i) $\lim_{x \rightarrow 0} \left(\frac{(1/2)^x - 1}{x} \right)$

j) $\lim_{x \rightarrow 0^+} \sqrt{x} \ln x$

k) $\lim_{x \rightarrow \infty} x \sin \frac{1}{x}$

l) $\lim_{x \rightarrow 1^+} \left(\frac{1}{x-1} - \frac{1}{\ln x} \right)$

m) $\lim_{x \rightarrow 0} \left(\operatorname{cosec} x - \frac{1}{x} \right)$

n) $\lim_{x \rightarrow 1^+} x^{1/(1-x)}$

o) $\lim_{x \rightarrow \infty} \left(\frac{x+2}{x-1} \right)^x$



Faculty of computer science and information technology

Year 2018/2019

Level:1

Course Name: Calculus 1

Course Code: MAT102

Sheet No.:6

Semester: Fall 2018

Instructor: Dr. RawhyIsmail

Problem 1)

i) Find the **vertical** asymptotes of the graph of each of the following functions:

$$i) y = \left(\frac{2x - 1}{x^2 - 7x - 8} \right)$$

$$ii) y = \left(\frac{x^2 - x - 2}{x^2 + 1} \right)$$

$$iii) y = \left(\frac{e^x}{x^3 - 4x} \right)$$

$$v) y = x \ln x$$

ii) Find the **horizontal** asymptotes of the graph of each of the following functions:

$$i) y = \left(\frac{3x^2 + 7x - 9}{x^2 - 4} \right)$$

$$ii) y = \left(\frac{2\sqrt{x}}{3x + 1} \right)$$

$$iii) y = \left(\left(\frac{1}{x} \right)^{1/x} \right)$$

Problem 2) Graph the following functions:

$$i) y = \left(\frac{2x^2}{x^2 - 1} \right)$$

$$ii) y = \left(\frac{3}{x^2 + 5x} \right)$$

$$iii) y = \left(\frac{x^2 + 5x - 6}{x^2} \right)$$

Problem 3)

Find a linear approximation of $f(x)$ near the point x_0 and hence approximate the value of $f(x)$ at $x = a$ and give the error:

$$i) f(x) = xe^{-x} \quad x_0 = 0 \quad a = 0.003$$

$$ii) f(x) = \cosh x^2 \quad x_0 = 1 \quad a = 0.998$$

Problem 4)

The radius of a sphere is measured to be 10 cm. Due to error in measuring; there is a maximum possible error of 0.001 cm in measuring the radius. Find upper bounds for the errors in calculating both the volume and the surface area of the sphere.